Design and Development of a 250g-class 3-inch Nano FPV Quadcopter

Executive Summary

This report documents the end-to-end engineering of a **250g-class (Nano) 3-inch FPV quadcopter**, emphasizing: (i) low-mass, high-thrust propulsion; (ii) robust electrical integration; (iii) a **custom flight-controller (FC)** centered on an STM32H743-class MCU and a vibration-tolerant sensor stack (**BMI088 IMU**, **BMP388 barometer**, **IST8310 magnetometer**); (iv) high-quality RC control/telemetry via **ExpressLRS**; and (v) navigation safety features using a **u-blox M10-class GNSS** and optional ToF ranging (VL53L1X).

Primary compliance and operational framing is taken from India's **Drone Rules, 2021**, published under the Ministry of Civil Aviation[1] and enforced by the Directorate General of Civil Aviation[2], including the "Nano" weight class and license exemptions. [3] The rules repeatedly reference the DGCA Digital Sky Platform[4] for licensing/registration workflows and related operational controls (where applicable). [3]

The core avionics design uses manufacturer primary documentation from STMicroelectronics[5] for the STM32H743/753 family (Cortex-M7 up to 480 MHz). [6] Sensors are justified using datasheets from Bosch Sensortec[7] (BMI088 IMU, BMP388 barometer) and iSentek[8] (IST8310 magnetometer, I2C output up to 400 kHz plus self-test/temperature compensation). [9] GNSS selection is anchored to u-blox[10] M10 references (MAX-M10S), supporting concurrent multi-constellation L1 reception (GPS/GLONASS/Galileo/BeiDou, configuration-dependent). [11]

The radio stack is built around ExpressLRS[12] using CRSF over UART with distinct control and telemetry lines. [13] Firmware options are evaluated using official docs for Betaflight[14] (DShot, ESC telemetry, GPS Rescue limitations) and ArduPilot[15] (EKF sensor fusion; Loiter and RTL behavior/requirements). [16]

For procurement, the report provides a supplier mapping for India-centric ecosystems (e.g., Robu.in[17] and RC Mumbai[18]) plus global sourcing via GetFPV[19] and electronics channels like Mouser Electronics[20] / DigiKey[21]. [22]

All worked examples use the user-specified baseline **AUW = 208 g** and both **4S 650 mAh** and **4S 850 mAh** battery options, with explicitly stated assumptions where measured motor/prop thrust curves are unavailable.

Objectives and Target Specifications

Project objectives

The project is structured to deliver a submission/portfolio-ready system, not just a flying prototype:

1. **Build a Nano-class 3-inch FPV quadcopter** that can remain under the **≤250 g "Nano" threshold (AUW including payload)** under defined configurations. [23]
2. **Demonstrate computation-backed design choices** (propulsion margin, battery sizing, and power budget) with unit-consistent formulas and sensitivity cases.
3. **Develop and document a custom FC architecture** using STM32H743-class performance and a sensor suite appropriate for vibration-heavy multirotor dynamics. [24]
4. **Support safety-oriented navigation modes**, ranging from Betaflight GPS Rescue (emergency return-to-range) to ArduPilot Loiter/RTL (true navigation behaviors), with clear limitations and test procedures. [25]

Target specifications

India's Drone Rules draft text explicitly defines "Nano" as **≤250 g AUW including payload**, which is the controlling top-level requirement for mass budgeting. [23] The rules also state **no remote pilot licence is required for operating a nano unmanned aircraft system**. [26]

Attribute	Target	Notes / verification method
AUW (baseline)	208 g	Given baseline for worked examples; must be verified on scale with the exact config.
AUW (compliance cap)	≤ 250 g	Nano class threshold (AUW including payload). [23]
Payload margin (design intent)	0–40 g (config-dependent)	"Payload" is treated as added mass above baseline; must retain ≤250 g cap. [23]
Thrust-to-weight (T/W)	≥ 4:1 (goal), ≥ 3:1 (minimum practical)	Enables control authority and safety margins; validated via thrust test + current logging.
Hover time (650 mAh)	scenario-based 4–12 min	Computed from usable Wh and average power; validated by logs and landing voltage.
Hover time (850 mAh)	scenario-based 5–15 min	Higher Wh but higher AUW; may reduce agility.
Max speed	pitch-limited estimate 40–70 km/h	Requires prop pitch and loaded RPM; treated as estimate unless GPS/blackbox verified.

Attribute	Target	Notes / verification method
RC link	ExpressLRS CRSF UART	Requires two pins: control + telemetry. [13]
Navigation	u-blox M10-class GNSS; optional ToF	Multi-constellation improves availability; ToF assists near-ground ranging. [27]

Engineering assumptions used consistently

Parameter	Value	Rationale
Gravity, g	9.80665 m/s^2	Standard gravity.
Air density, ρ	1.18 kg/m^3	Assumed warm low-altitude conditions (explicit assumption).
Prop diameter, D	3 in = 76.2 mm = 0.0762 m	3-inch class prop.
Battery nominal voltage	4S ≈ 14.8 V	LiPo nominal 3.7 V/cell ×4 (explicit assumption).
Usable energy fraction	80%	Conservative "usable Wh" assumption for LiPo longevity/safety (explicit assumption).
Static thrust modeling coefficient	$C_T = 0.10$ (illustrative)	Used only to show scaling when measured curves are missing (explicit assumption).

System Design and Engineering Calculations

Mechanical design rationale

A 3-inch quad is dominated by "small multirotor constraints": vibration, wiring density, CG sensitivity, and component packaging.

A representative 3-inch carbon frame spec (HF150, 150 mm wheelbase) cites ~40 g frame weight and carbon plate thicknesses (arms ~4 mm, plates ~2 mm). [28] This frame class provides sufficient stiffness for 1404 motors while keeping AUW low.

CG and mass distribution
For attitude control, keeping the center of gravity near the geometric center minimizes trim offsets and reduces required steady-state control torque. A practical engineering measure is the axis moment contribution:

$$I \approx \sum_i m_i r_i^2$$

where m_i is component mass and r_i is its distance from the axis. Batteries should be centered because they dominate m_i in Nano builds.

Electrical architecture overview

A clean architecture separates high-current power distribution from sensitive sensing:

1. **Battery (XT30)** → **4-in-1 ESC (VBAT)** → motors
2. ESC BEC / or FC regulator subtree → **5 V rail** (receiver, GNSS module, ToF module, FPV electronics if compatible)
3. **3.3 V rail** → MCU + sensors (IMU, baro, magnetometer, flash)

For command signaling, DShot (digital ESC protocol) supports telemetry features; Betaflight documents DShot timing/telemetry behavior and also describes ESC telemetry methods (DShot telemetry vs serial ESC telemetry). [29]

Propulsion calculations

Baseline hover thrust requirement (worked example)

Given: - AUW mass $m = 208$ g $= 0.208$ kg - $g = 9.80665$ m/s^2

Total thrust required for hover:

$$T_{\text{total,hover}} \approx mg = (0.208)(9.80665) = 2.0398 \text{ N}$$

Per-motor hover thrust (quad, 4 motors):

$$T_{\text{per,hover}} = \frac{T_{\text{total,hover}}}{4} = \frac{2.0398}{4} = 0.5099 \text{ N}$$

Convert to "grams-force equivalent" (useful in FPV sizing):

$$1 \text{ N} \approx 101.97 \text{ gf} \Rightarrow 0.5099 \text{ N} \approx 52 \text{ gf}$$

So each motor needs only ~**52 g** static thrust to hover this 208 g craft; robust designs target **>200 g per motor** to achieve ≥4:1 T/W.

Thrust-to-weight sizing targets (worked example)

Weight in grams-force for sizing is ~208 gf. Therefore:

- For **4:1 T/W**, total thrust target:
  $T_{\text{total}} \approx 4 \times 208 = 832 \text{ gf} \Rightarrow T_{\text{per motor}} \approx 208 \text{ gf}$
- For **5:1 T/W**, per motor target:
  $T_{\text{per motor}} \approx \frac{5 \times 208}{4} = 260 \text{ gf}$

This establishes the minimum thrust capability your motor/prop must demonstrate on a thrust stand at acceptable current draw.

Motor selection linkage to current limits

A representative 1404 motor choice from the project notes is the iFlight XING2 1404 family. The manufacturer store listing includes **3S–4S input**, **~9.1 g weight (with wire)** and provides example **peak current** values for KV variants (e.g., ~15.84 A class). [30] These values are essential for ESC sizing and battery discharge calculations.

A representative ESC choice in the notes is the T-Motor Mini F45A 4-in-1 (45 A class), commonly advertised with protocol support including DShot and higher-voltage capability (up to 6S, depending on SKU). [31]

Static thrust curve (modeled example with explicit assumptions)

If a measured thrust curve for the exact motor+prop is unavailable, a first-pass scaling model uses:

$$T = C_T \rho n^2 D^4$$

where: - C_T is thrust coefficient (prop-dependent; **assumed here only**) - $\rho = 1.18$ kg/m^3 (assumed) - $n = \text{RPM}/60$ - $D = 0.0762$ m

Assumption: $C_T = 0.10$ (illustrative only; must be replaced by measured data).

Modeled thrust per motor:

RPM	n (rev/s)	Thrust T (N)	Thrust (gf)
20,000	333.3	0.442	45
30,000	500.0	0.995	101
40,000	666.7	1.768	180
50,000	833.3	2.763	282
60,000	1000.0	3.978	406

Interpretation: This modeled curve suggests that if the propulsion system can sustain ~50k loaded RPM with the assumed coefficient, a per-motor thrust on the order of ~280 g could be possible—enough for ≥5:1 T/W at 208 g AUW. This is **not** a performance claim; it shows the scaling behavior and why thrust-stand tests are mandatory to replace C_T with real data.

How to obtain real thrust curves (required for final validation): - Use a thrust stand with calibrated load cell. - Record thrust (N), current (A), voltage (V), RPM (if available via ESC telemetry), and motor temperature. - Sweep throttle in steps (e.g., 10% increments), holding each step until readings stabilize.

Betaflight's DShot and ESC telemetry documentation supports the concept of extracting RPM/telemetry data pathways (where supported by ESC/firmware). [29]

Induced power (actuator-disk) calculation

This is a **lower-bound** hover power model (ideal induced power only). It will underpredict real electrical power because it excludes prop profile power, frame interference, motor/ESC losses, and non-idealities.

Per rotor disk area:

$$A = \pi\left(\frac{D}{2}\right)^2 = \pi(0.0381)^2 = 0.00456\ \mathrm{m}^2$$

Per-motor hover thrust was computed above:

$$T_{\text{per,hover}} = 0.5099\ \mathrm{N}$$

Induced velocity (hover):

$$v_i = \sqrt{\frac{T}{2\rho A}} = \sqrt{\frac{0.5099}{2(1.18)(0.00456)}} = 6.88\ \mathrm{m/s}$$

Ideal induced power per motor:

$$P_{i,\text{per}} = Tv_i = (0.5099)(6.88) = 3.51\ \mathrm{W}$$

Total ideal induced power (4 motors):

$$P_{i,\text{total}} \approx 4 \times 3.51 = 14.0\ \mathrm{W}$$

Engineering use: treat 14 W as a strict aerodynamic lower bound. Real electrical hover power for a 3-inch nano is typically several times higher; therefore, flight time must be predicted from logged electrical power rather than induced power alone.

Battery energy and flight time calculations (worked examples)

Assume nominal 4S voltage:

$$V_{\text{nom}} = 14.8\ \mathrm{V}$$

Battery energy in watt-hours:

$$E_{\text{Wh}} = V_{\text{nom}} \times \text{Capacity(Ah)}$$

Usable energy (80% usable assumption):

$$E_{\text{usable}} = 0.8E_{\text{Wh}}$$

4S 650 mAh case

$$\text{Capacity} = 650\ \text{mAh} = 0.65\ \text{Ah}$$

$$E_{\text{Wh}} = 14.8 \times 0.65 = 9.62\ \text{Wh}$$

$$E_{\text{usable}} = 0.8 \times 9.62 = 7.70 \text{ Wh}$$

4S 850 mAh case

$$\text{Capacity} = 850 \text{ mAh} = 0.85 \text{ Ah}$$

$$E_{\text{Wh}} = 14.8 \times 0.85 = 12.58 \text{ Wh}$$

$$E_{\text{usable}} = 0.8 \times 12.58 = 10.06 \text{ Wh}$$

Flight time estimate from average power

$$t = \frac{E_{\text{usable}}}{P_{\text{avg}}}$$

Because P_{avg} depends on tune, prop choice, wind, and flight style, estimate multiple scenarios:

Battery	E_{usable} (Wh)	P_{avg} (W)	t (min)
4S 650 mAh	7.70	40	11.5
4S 650 mAh	7.70	60	7.7
4S 650 mAh	7.70	90	5.1
4S 650 mAh	7.70	120	3.8
4S 850 mAh	10.06	40	15.1
4S 850 mAh	10.06	60	10.1
4S 850 mAh	10.06	90	6.7
4S 850 mAh	10.06	120	5.0

Interpretation: 850 mAh increases usable energy by ~31% vs 650 mAh, but also increases AUW, which increases hover thrust and likely increases P_{avg}. The "best battery" must therefore be chosen by measured logs, not Wh alone.

C-rating discharge math (worked examples)

$$I_{\max} = C \times \text{Capacity(Ah)}$$

For 650 mAh (0.65 Ah): - 50C → $I_{\max} = 32.5$ A
- 75C → $I_{\max} = 48.8$ A
- 95C → $I_{\max} = 61.8$ A

For 850 mAh (0.85 Ah): - 50C → $I_{\max} = 42.5$ A
- 75C → $I_{\max} = 63.8$ A
- 95C → $I_{\max} = 80.8$ A

Engineering caution: mode transitions (punch-outs) can create burst currents. Motor peak-current guidance from the XING2 listing provides a basis to check whether 4 motors at peak could exceed realistic battery capability. [32]

Max speed estimate (pitch-speed upper bound)

Pitch-speed upper bound (ignores slip and drag):

$$V_{\text{pitch}} \approx \text{pitch} \cdot \frac{\text{RPM}}{60}$$

If a 3-inch prop has pitch $p = 0.9$ in $= 0.02286$ m (assumed—must be confirmed from your exact prop SKU), then: - 40k RPM → $V = 0.02286 \cdot 40000/60 = 15.24$ m/s $= 54.9$ km/h - 50k RPM → $V = 19.05$ m/s $= 68.6$ km/h

In practice, slip + aerodynamic drag reduce achievable speed below pitch speed, so GPS/blackbox measurement should be used for a defensible "max speed" claim.

Flight Controller and Software Architecture

Custom FC hardware blocks and interfaces

The MCU baseline is STM32H743/753 (Arm Cortex-M7 up to 480 MHz), providing headroom for high-rate control loops, filtering, logging, and peripheral IO. [6]

Sensor suite: - **BMI088**: 6-DoF IMU combining a triaxial gyroscope and triaxial accelerometer. [33]
- **BMP388**: digital pressure sensor specifically positioned for drone altitude tracking (temperature and pressure measurement). [34]
- **IST8310**: 3-axis digital magnetometer with I2C output (fast mode up to 400 kHz) and built-in temp compensation/self-test. [35]
- **GNSS (M10-class)**: multi-constellation L1 support with concurrent reception (configuration-dependent), improving satellite availability in challenging environments. [11]
- **VL53L1X ToF**: miniature ToF ranging sensor with accurate ranging up to ~4 m and ranging frequency up to ~50 Hz, suitable for near-ground height/ranging augmentation (not a full autonomy sensor by itself). [36]

FC system block diagram (Mermaid)

```
flowchart LR
  VBAT[4S LiPo VBAT] --> ESC[4-in-1 ESC]
  ESC --> M1[Motor 1]
  ESC --> M2[Motor 2]
  ESC --> M3[Motor 3]
  ESC --> M4[Motor 4]

  subgraph FC[Custom Flight Controller]
    MCU[STM32H743-class MCU]
```

```
    IMU[BMI088 IMU]
    BARO[BMP388 Barometer]
    MAG[IST8310 Magnetometer]
    FLASH[Blackbox Flash (optional)]
    REG5[5V Regulator]
    REG33[3.3V Regulator]
  end

  MCU -->|DShot| ESC
  REG5 --> RX[ExpressLRS RX]
  MCU <-->|CRSF UART| RX
  MCU <-->|UART| GPS[GNSS (u-blox M10-class)]
  MCU <-->|I2C| TOF[VL53L1X ToF]
  MCU <-->|SPI or I2C| IMU
  MCU <-->|I2C or SPI| BARO
  MCU <-->|I2C| MAG
  MCU --> FLASH
```

[37]

PCB design considerations (submission-quality notes)

A Nano FC board must handle simultaneous requirements: low-noise sensing and high di/dt power switching nearby.

Key design practices: - **4-layer PCB recommended**: continuous ground plane, controlled return paths for high-speed SPI, and isolation between switching power and sensors (assumption based on standard embedded layout practice). - **IMU placement**: near the board CG; shortest possible SPI traces; avoid placing the IMU over cutouts or near mounting holes that transmit vibration. - **Power integrity**: local decoupling per MCU power pin, LC filtering where needed for analog rails, and careful separation of 3.3V sensor rail from 5V switching ripple. - **Magnetometer placement**: magnetometers are exceptionally sensitive to motor current loops; either place it on a GPS mast module or locate it as far from VBAT and motor phase wires as possible. ArduPilot explicitly notes that low magnetic interference and low vibration are important for good Loiter performance. [38]

Logical pin/interface mapping table (MCU pin numbers intentionally unspecified)

This table maps *logical buses*; the exact MCU pins depend on your STM32 package and PCB routing and must be defined in the schematic/firmware target definition.

Peripheral	Interface	Logical port	Electrical level	Notes
ESC (4 motors)	DShot	TIM outputs	3.3V logic typical	DShot timing/telemetry described in Betaflight

Peripheral	Interface	Logical port	Electrical level	Notes
				docs. [39]
ExpressLRS receiver	UART (CRSF)	UART_RX/UART_TX	3.3V (verify RX)	UART-based ELRS uses CRSF; uses separate control + telemetry lines. [13]
GNSS module	UART	UART_GPS	3.3V or 5V-tolerant (module dependent)	u-blox M10 modules provide multi-GNSS capability. [40]
BMI088 IMU	SPI preferred	SPI_IMU	3.3V	IMU is a 6DoF device; SPI reduces latency/bus contention. [33]
BMP388 barometer	I2C or SPI	I2C_SENS or SPI_SENS	3.3V	Supports digital interfaces; drone altitude tracking focus. [41]
IST8310 magnetometer	I2C	I2C_SENS	3.3V	I2C up to 400 kHz; includes self-test/temp comp. [35]
VL53L1X ToF	I2C	I2C_SENS	3.3V (often 5V-tolerant modules)	Up to ~4 m, ~50 Hz in datasheet. [36]
Blackbox flash (optional)	QSPI/SPI	SPI_FLASH/QSPI	3.3V	Recommended for tuning and verification logs (assumption).

Firmware options and trade-offs

Betaflight path (FPV-first)
Betaflight is optimized for acro/rate flight and supports DShot and ESC telemetry workflows. [29] If navigation safety is added, GPS Rescue is described as an emergency feature to bring the quad back into range and is explicitly "not meant to be a reliable Return to Home mode." [42] This affects how GPS is validated and how claims are phrased in an academic submission.

ArduPilot path (navigation-first)
ArduPilot uses an EKF to estimate position/velocity/orientation using IMU, compass, GPS, and barometric pressure (and other sensors when present). [43] It provides true navigation modes such as Loiter (position hold) and RTL (return to launch), with documented

dependence on good GPS lock, low magnetic interference, and low vibration. [44] This aligns naturally with the inclusion of IST8310 + u-blox M10 + BMP388.

Custom RTOS / custom stack
A custom stack is feasible on STM32H743-class compute, but the engineering cost is high: sensor drivers, deterministic scheduling, calibration workflows, fault handling, and a safe flight test campaign. STM32H743-class performance is documented at up to 480 MHz Cortex-M7. [45]

Sensor fusion and control equations

PID control (axis-level)

$$u(t) = K_p e(t) + K_i \int_0^t e(\tau)\, d\tau + K_d \frac{de(t)}{dt}$$

Where $e(t)$ is attitude/rate error and $u(t)$ drives motor mixing.

EKF overview (autonomy-grade estimation)
ArduPilot documents EKF-based estimation of position/velocity/orientation using gyroscopes, accelerometers, compass, GPS, and barometric pressure measurements. [43]

Sampling rates (engineering guidance with explicit uncertainty)
Betaflight and ArduPilot loop rates depend on configuration and CPU load; STM32H743-class clock headroom (480 MHz) supports high-rate loops, but exact values must be measured (CPU utilization, scheduler timing) on the final firmware target. [45]

Bill of Materials and Costing

Supplier directory (codes used in tables)

To avoid repeating supplier names, codes are used below:

- **IN-A**: Robu.in [46]
- **IN-B**: RC Mumbai [47]
- **IN-C**: Robokits (India distributor example for ESC listing) [48]
- **GL-A**: GetFPV [49]
- **GL-B**: iFlight official store listing [32]
- **GL-C**: T-Motor Hobby (manufacturer store) [50]
- **GL-D**: ST datasheet portal (VL53L1X) [51]

For IC procurement, Mouser/DigiKey are typical global channels; part availability varies by date and region. [52]

Detailed BOM table (core airframe + avionics)

Subsystem	Item	Example part / model	Qty	Preferred suppliers	Primary datasheet / official reference
Frame	3" carbon frame	HF150 150 mm 3-inch (40 g class)	1	IN-A	Frame spec listing. [28]
Motors	1404 class motors	iFlight XING2 1404 3800KV (3S–4S, ~9.1 g w/ wire)	4	GL-B, IN-A, GL-A	XING2 1404 specs include KV options, mass, voltage range and peak current guidance. [53]
ESC	4-in-1 ESC	T-Motor Mini F45A 4-in-1 (45 A class)	1	IN-A, IN-C, GL-A, GL-C	Example distributor/manufacturer listings for rating/protocol support. [54]
Propellers	3-inch props	Gemfan 75 mm props (exact SKU TBD)	1–2 sets	IN-B	Prop options and size availability. [55]
Battery	LiPo (option A)	4S 650 mAh	1	IN/GL market-dependent	Used in energy calculations (assumption).
Battery	LiPo (option B)	4S 850 mAh	1	IN/GL market-dependent	Used in energy calculations (assumption).
Flight controller	Custom FC PCB	STM32H743-class	1	Electronics distributors	STM32H743/753 family overview and documentation page. [6]
IMU	6DoF IMU	BMI088	1	Electronics distributors	BMI088 datasheet. [33]
Barometer	Pressure sensor	BMP388	1	Electronics distributors	BMP388 datasheet / drone suitability notes. [56]

Subsystem	Item	Example part / model	Qty	Preferred suppliers	Primary datasheet / official reference
Magnetometer	3-axis mag	IST8310	1	Electronics distributors	IST8310 datasheet (I2C fast mode, temp comp, self-test). [35]
GNSS	GPS module	u-blox M10-class module (MAX-M10S reference)	1	Electronics distributors / FPV GNSS modules	u-blox M10 datasheet + integration manual. [11]
RC receiver	RX	ExpressLRS UART receiver (e.g., EP1-class)	1	FPV suppliers	ELRS receiver wiring and CRSF requirements. [13]
ToF ranging	Height/ranging aid	VL53L1X module	1	IN/GL electronics	VL53L1X datasheet (4 m, 50 Hz). [36]
Telemetry/logging	ESC telemetry	DShot/telemetry support	—	—	Betaflight ESC telemetry

Cost breakdown (method + partially sourced examples)

Prices vary continuously and should be captured in an appendix with date-stamped screenshots/POs. Where a live listing is available in sources, it is cited; otherwise, the line is "TBD."

Item	Qty	Unit cost (INR)	Subtotal (INR)	Source / notes
XING2 1404 motor	4	(listing-based)	—	Example India listing includes price at time of capture. [58]
HF150 frame	1	TBD	TBD	Frame spec referenced; capture current price during procurement. [28]
T-Motor Mini F45A 4-in-1 ESC	1	TBD	TBD	Multiple supplier listings; capture a single selected SKU price. [59]
GNSS module	1	TBD	TBD	Module selection depends on FPV packaging; u-blox datasheets used for

Item	Qty	Unit cost (INR)	Subtotal (INR)	Source / notes
				capability. [40]
Sensors (BMI088/BMP388/IST8310)	1 set	TBD	TBD	Source via manufacturer datasheets and electronics distributors. [9]
Electronics fabrication (custom FC PCB)	1–N	TBD	TBD	Dependent on layers, finish, stencil, assembly method (explicit project variable).
Tools/consumables	—	TBD	TBD	Solder, wires, heatshrink, fasteners, TPU prints, adhesives (explicit project variable).

Recommendation for academic submission: include a procurement appendix that tabulates the final spend, and attribute each line to invoice/PO evidence.

Integration and Assembly Procedure

Schematic-level wiring diagram (reference)

This is a **logical connectivity** diagram suitable for a report. Pad names differ by FC/ESC PCB.

```
Battery (4S LiPo)
  ├── XT30 (+VBAT) ────────────────┐
  └── XT30 (GND)  ─────────────────┘
                  │
                  ▼
            4-in-1 ESC (VBAT in)
              ├── Motor 1 (3-phase)
              ├── Motor 2 (3-phase)
              ├── Motor 3 (3-phase)
              └── Motor 4 (3-phase)

Custom Flight Controller
  Power:
    ├── VBAT sense (optional from ESC VBAT)
    ├── 5V regulator → ELRS RX + GNSS + ToF (+ optional FPV)
    └── 3.3V regulator → MCU + IMU + baro + magnetometer + flash

  Signals:
    ├── DShot1..4 → ESC signal pads 1..4
    ├── UART_Rx/Tx (CRSF) ↔ ExpressLRS RX
    ├── UART (GNSS) ↔ GPS module
```

```
    └── I2C bus ↔ BMP388 + IST8310 + VL53L1X (if addressed and pull-ups sized
correctly)
```

DShot and ESC telemetry capabilities are documented in Betaflight sources; ExpressLRS receiver wiring uses CRSF over UART with separate control and telemetry signals. [60]

Assembly steps (high-yield sequencing)

1. **Dry-fit** the stack (ESC + FC) and confirm 20×20 mounting, cable clearance, and correct motor screw length (avoid winding contact).
2. Install ESC; solder battery pigtail; ensure strain relief; add input capacitance if required by your ESC vendor recommendations (project-specific).
3. Install motors; solder motor phases; ensure consistent motor-wire lengths to reduce imbalance in resistance/EMI.
4. Install FC; ensure IMU orientation is correct in firmware; soft-mount if the mechanical stack transmits vibration.
5. Install ExpressLRS receiver; wire UART as CRSF; verify "Serial RX" configuration in FC software. [13]
6. GNSS: mount on a mast or rear deck far from VBAT and motor phase wires; consider magnetometer interference (especially in Loiter). [61]
7. Install ToF sensor downward-facing; ensure clear field of view; wire to I2C. [36]
8. Pre-power checks: continuity (VBAT to GND), smoke-stopper, then low-current power.
9. Firmware bring-up and motor direction validation with props removed; enable telemetry/logging where possible for later analysis. [62]

Verification and Testing Plan

A submission-grade validation plan must include staged tests with measurable pass criteria.

Test plan table

Stage	Test	Setup	Measurements	Pass criteria
Bench	Continuity & polarity	Multimeter	VBAT–GND resistance, 5V–GND	No short; correct polarity
Bench	Smoke-stopper power-up	Current-limited supply	Idle current, heat	No abnormal draw/heating
Bench	Sensor bring-up	Firmware sensor readout	gyro/accel/baro/mag noise	Readings stable; correct axes [9]
Bench	ELRS link	Radio + receiver	link stability, failsafe	CRSF data OK; failsafe works [13]
Bench	Motor test	DShot control	motor direction,	Correct motor mapping

Stage	Test	Setup	Measurements	Pass criteria
	(no props)		desync	[63]
Tether	Hover check	Tether rig	stability, oscillation, current	Stable hover; no runaway heating
Flight	First hover & land	Open area	motor temp, log review	Safe hover; acceptable temps
Flight	Flight time validation	Standardized profile	V(t), I(t), P(t)	Measured time aligns with Wh model
Flight	GPS safety mode	Controlled activation	behavior, recovery	For Betaflight: returns-to-range (not full RTH) [42]
Flight	Nav modes (ArduPilot)	Loiter + RTL	drift, heading, vibration	Good Loiter requires low mag interference and low vibration; RTL returns to launch [44]

Data capture and analysis expectations

- If using Betaflight: use ESC telemetry pathways where available to collect RPM/current proxies and to diagnose desync/overheating behavior. [64]
- If using ArduPilot: validate EKF health and sensor consistency; EKF fuses IMU, compass, GPS, and baro. [43]

Safety, Compliance, Risk, and Future Work

India-specific regulatory compliance

Nano classification: The Drone Rules draft text classifies drones by maximum all-up weight including payload and defines "Nano" as **≤250 g**. [23]
Remote pilot licence exemption: The gazette text states **no remote pilot licence is required for operating a nano unmanned aircraft system**. [26]
DigitalSky linkage: The rules reference the Digital Sky platform in the regulatory workflow context; therefore, document-checking and operational procedures should align with DigitalSky processes where applicable. [3]

Operational caution: This report is not legal advice. For a submission, include a compliance appendix that restates the relevant clauses you relied upon and your operating constraints (area selection, permissions where applicable, safety perimeter, and emergency procedures). [3]

Risk analysis

Risk	Failure mode	Likely causes	Controls / mitigations
LiPo thermal	Fire on	Overcharge, physical	Charge supervision,

Risk	Failure mode	Likely causes	Controls / mitigations
event	charge/crash	damage, short	storage protocol, preflight inspection
Flyaway	Loss of control/video	link issues, failsafe misconfig	Verify CRSF wiring, failsafe tests, telemetry monitoring [13]
“False RTH” expectation	Unsafe reliance on GPS rescue	misunderstanding feature scope	Betaflight GPS Rescue is not full reliable RTH; treat as emergency only [42]
Poor Loiter/RTL	Drift or instability	vibration, compass interference	Mitigate vibration and magnetic interference; required for good Loiter [38]
ESC overheat/desync	Motor stop mid-flight	excessive current, wrong settings	ESC headroom, telemetry monitoring, prop choice validation [62]
Sensor rail noise	EKF/PID instability	poor grounding, regulator ripple	4-layer PCB, clean return paths, decoupling, rail filtering (design practice assumption)

Development timeline

Phase	Deliverables	Duration (typical)
Requirements & rules mapping	Specs + Nano compliance plan	1 week
Procurement	Parts list, supplier selection, purchase	1–3 weeks
Mechanical integration	CAD/TPU mounts, CG plan	1–2 weeks
Electrical integration	Wiring harness, smoke-stopper pass	1 week
Firmware bring-up	ELRS + sensors + motor mapping	1 week
Flight tuning	Stable tune + thermal validation	1–2 weeks
Navigation validation	GPS Rescue (BF) or Loiter/RTL (AP)	1–2 weeks
Final verification & report	Logs + plots + appendix	1 week

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flowchart TD
  A[Requirements and Nano compliance mapping] --> B[Procurement]
  B --> C[Mechanical integration and CG check]
  C --> D[Electrical integration and bench safety tests]
  D --> E[Firmware bring-up and motor mapping]
  E --> F[Tethered hover and first flight]
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F --> G[Tuning, thermal checks, flight time validation]
G --> H[Navigation validation]
H --> I[Final report, evidence pack, build checklist]
```

Future improvements

1. **Custom FC iteration:** improved EMI isolation, IMU placement refinements, and expanded logging to support repeatable tuning. STM32H743-class headroom supports richer logging and filtering, if PCB and firmware are designed for it. [45]
2. **Autonomy upgrades:** migrate to ArduPilot Loiter/RTL and waypoint missions where the EKF and navigation stack are central requirements. [65]
3. **ToF-assisted landing:** use VL53L1X range data (up to ~4 m and ~50 Hz) to improve low-altitude height estimation (fusion strategy must be validated). [51]
4. **AI/vision pipeline (optional):** add a companion compute module (weight/power permitting) for vision-based obstacle avoidance; this likely pushes AUW close to or beyond Nano limits and must be treated as a separate compliance configuration. [23]

Recommendations and concise build checklist

Recommendations (engineering)
Maintain Nano headroom by defining configuration sets (e.g., “650 mAh acro,” “850 mAh endurance,” “GNSS+ToF nav”) and measuring AUW for each against the ≤250 g cap. [23]
Treat Betaflight GPS Rescue as emergency return-to-range rather than guaranteed RTH; use ArduPilot if true navigation behavior is a core project claim. [66]

Concise build checklist (submission-ready) - AUW measured and documented for each configuration; Nano compliance checked (≤250 g AUW incl. payload). [23]
- No remote pilot licence requirement verified for Nano in India (documentation cited in appendix). [26]
- Power wiring polarity verified; smoke-stopper pass recorded.
- Motors mapped and directions correct under DShot; telemetry/logging enabled where supported. [39]
- ExpressLRS CRSF UART wired correctly (control + telemetry lines); failsafe validated. [13]
- GNSS receives satellites; magnetometer interference assessed (especially before Loiter). [67]
- Flight time measured and compared to Wh model for both 650 mAh and 850 mAh batteries (include plots/log summaries).
- Navigation feature testing documented: - Betaflight: GPS Rescue behavior demonstrated with limitation statement. [42]
- ArduPilot: demonstrate Loiter + RTL and document EKF health. [68]

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[1] [6] [24] [37] [45] STM32H743/753 Arm Cortex-M7 Microcontrollers (MCUs)

https://www.st.com/en/microcontrollers-microprocessors/stm32h743-753.html?utm_source=chatgpt.com

[2] [31] [48] [54] T-Motor Mini F45A 6S 4IN1 32bit

https://robokits.co.in/multirotor-spare-parts/tiger-motor-esc-and-propeller/t-motor-esc/fpv-series/t-motor-mini-f45a-6s-4in1-32bit?srsltid=AfmBOoqbLKy8LHThP528twb7ucrBJsBqSJD-oAi53x1CUXgLcTHTfXf8&utm_source=chatgpt.com

[3] [8] [10] [26] नागर जिमानन मंत्रालय अजधसूचना नई दिल्ली , 25 अगस्ट

https://egazette.gov.in/WriteReadData/2021/229221.pdf?utm_source=chatgpt.com

[4] [57] [62] [64] ESC Telemetry

https://betaflight.com/docs/wiki/guides/current/ESC-Telemetry?utm_source=chatgpt.com

[5] [9] [33] BMI088 Datasheet

https://www.bosch-sensortec.com/media/boschsensortec/downloads/datasheets/bst-bmi088-ds001.pdf?utm_source=chatgpt.com

[7] [34] BMP388 Digital, barometric pressure sensor

https://www.bosch-sensortec.com/media/boschsensortec/downloads/product_flyer/bst-bmp388-fl000.pdf?utm_source=chatgpt.com

[11] [27] [40] MAX-M10S Data sheet

https://content.u-blox.com/sites/default/files/MAX-M10S_DataSheet_UBX-20035208.pdf?utm_source=chatgpt.com

[12] [35] IST8310 3D Magnetometer Datasheet

https://isentek.com/userfiles/files/IST8310Datasheet_3DMagneticSensors.pdf?utm_source=chatgpt.com

[13] [18] [21] Receiver Wiring

https://www.expresslrs.org/quick-start/receivers/wiring-up/?utm_source=chatgpt.com

[14] [36] [51] Datasheet - VL53L1X

https://www.st.com/resource/en/datasheet/vl53l1x.pdf?utm_source=chatgpt.com

[15] [59] T Motor MINI F45A 6S 4IN1

https://robu.in/product/t-motor-mini-f45a-6s-4in1/?utm_source=chatgpt.com

[16] [29] [39] [60] [63] DSHOT

https://betaflight.com/docs/development/Dshot?utm_source=chatgpt.com

[17] [25] [42] [66] GPS Rescue before V4.1

https://betaflight.com/docs/wiki/guides/archive/GPS-Rescue-Mode?utm_source=chatgpt.com

[19] [23] to be published in the gazette of india, extraordinary, part ii

https://digitalsky.dgca.gov.in/assets/files/Draft-DronesRules-2021_Gazette%20version_14-Jul-2021.pdf?utm_source=chatgpt.com

[20] [38] [44] [61] [68] Loiter Mode — Copter documentation

https://ardupilot.org/copter/docs/loiter-mode.html?utm_source=chatgpt.com

[22] [28] [46] HF150 150mm 3 inch Carbon Fiber Frame

https://robu.in/product/hf150-150mm-3-inch-carbon-fiber-frame/?utm_source=chatgpt.com

[30] [32] [53] XING2 1404 Toothpick Ultralight Build (unibell)

https://shop.iflight.com/xing2-1404-toothpick-ultralight-build-unibell-pro1482?utm_source=chatgpt.com

[41] [56] BMP388 Digital pressure sensor

https://www.bosch-sensortec.com/media/boschsensortec/downloads/datasheets/bst-bmp388-ds001.pdf?utm_source=chatgpt.com

[43] [65] Extended Kalman Filter (EKF) — Copter documentation

https://ardupilot.org/copter/docs/common-apm-navigation-extended-kalman-filter-overview.html?utm_source=chatgpt.com

[47] GEMFAN 75MM DUCTED PROPS PC 3-BLADE

https://rcmumbai.com/products/gemfan-75mm-ducted-props-pc-3-blade?srsltid=AfmBOoqoKUCMNrdVdjNVRi-pH1ajJl2JBqIx_KwiuDXn43l2_vpGC9cT&utm_source=chatgpt.com

[49] iFlight XING2 1404 3000KV/3800KV/4600KV Toothpick ...

https://www.getfpv.com/iflight-xing2-1404-3000kv-3800kv-4600kv-toothpick-motor.html?srsltid=AfmBOop-1Y68UKSQM0ZSIz81DCnAwOlFGwUJ5PcOk5_RIhoG4-ySyEhv&utm_source=chatgpt.com

[50] T-MOTORHOBBY MINI F45A+MINI F7 20x20 Stack 3-6S ...

https://tmotorhobby.com/goods-1020-TMOTOR%2BMINI%2BF45AMINI%2BF7%2B20x20%2BStack%2B3-6S%2BBLHeli_32%2B4in1%2BFPV%2BDrones%2BESC.html?utm_source=chatgpt.com

[52] BMP388 - Mfr.: Bosch Sensortec

https://www.mouser.in/ProductDetail/Bosch-Sensortec/BMP388?qs=rrS6PyfT74eIcD5Rv1Kf7A%3D%3D&srsltid=AfmBOopDJaBsE9vQ3MWFoi_m8Wxg4c8-lQs8od-FMfaopXxnGnC_2fr-&utm_source=chatgpt.com

[55] By Size-Prop – Page 19

https://rcmumbai.com/collections/by-size-props?page=19&srsltid=AfmBOor9g5sj3_Kfv1cFLmBDO2H03r07FUvh3yQ_b12kWMYHijWSzZ4P&utm_source=chatgpt.com

[58] XING2 1404 FPV Motor Unibell - 3800KV

https://robu.in/product/xing2-1404-fpv-motor-unibell-3800kv/?utm_source=chatgpt.com

[67] MAX-M10S Integration manual

https://content.u-blox.com/sites/default/files/MAX-M10S_IntegrationManual_UBX-20053088.pdf?utm_source=chatgpt.com